

Computation

- Basic operations involving vectors and matrices (no matrix multiplication)
 - addition, subtraction, scalars, dot product, magnitude, angle.
- Find the reduced echelon form of a matrix.
- Find the general solution of the equation $A\mathbf{x} = \mathbf{b}$.
- Find the general solution of a system of equations from various information.
- Find the Cartesian equation of a hyperplane from various information.
- Find the normal vector to a hyperplane from various information.

Concept**Section 1.1**

- Algebraic computations and geometric interpretations of basic vector operations.
- Determine if a point is on a line.
- Find the parametric form/Cartesian equation of a line.

Section 1.2

- Compute dot product, angle, and magnitude of vectors.

Section 1.3

- Give Cartesian equation/general solution of a hyperplane.